How Ricardian Are We?

Jonathan J. Adams

Christian Matthes

T2M (HEC Montreal), June 2nd 2026

"The debts of a nation are debts due from the right hand to the left."

— David Ricardo (1817)

"The debts of a nation are debts due from the right hand to the left."

— David Ricardo (1817)

"The people who pay the taxes never so estimate them, and therefore do not manage their private affairs accordingly."

— David Ricardo (1820)

Question: How Ricardian Are Households?

- What share of future taxes do households save to pay for?

Question: How Ricardian Are Households?

- What share of future taxes do households save to pay for?
- Three other ways to think about it:
  - how much government debt households treat as net wealth?
  - how much of a debt-financed tax cut is offset by higher private saving?
  - the aggregate MPC out of debt-financed transfers?

Question: How Ricardian Are Households?

- What share of future taxes do households save to pay for?
- Three other ways to think about it:
  - how much government debt households treat as net wealth?
  - how much of a debt-financed tax cut is offset by higher private saving?
  - the aggregate MPC out of debt-financed transfers?
- The answer matters for almost any macroeconomic policy
  - fiscal policy, stabilization, debt management, monetary-fiscal interactions, etc.

Question: How Ricardian Are Households?

- What share of future taxes do households save to pay for?
- Three other ways to think about it:
  - how much government debt households treat as net wealth?
  - how much of a debt-financed tax cut is offset by higher private saving?
  - the aggregate MPC out of debt-financed transfers?
- The answer matters for almost any macroeconomic policy
  - fiscal policy, stabilization, debt management, monetary-fiscal interactions, etc.

Object of Interest

$$\theta = \text{share of future taxes households save to pay for}$$

Why Micro Evidence Is Not Enough

What Existing Micro Evidence Tells Us

How much consumption rises when a household receives more current resources.

This identifies the usual marginal propensity to consume (MPC).

What The Ricardian Question Needs

How much consumption rises after a debt-financed transfer or tax cut.

That is the MPC out of debt-financed transfers (MPC-DFT).

Why Micro Evidence Is Not Enough

What Existing Micro Evidence Tells Us

How much consumption rises when a household receives more current resources.

This identifies the usual marginal propensity to consume (MPC).

What The Ricardian Question Needs

How much consumption rises after a debt-financed transfer or tax cut.

That is the MPC out of debt-financed transfers (MPC-DFT).

- The missing intercept problem: micro evidence identifies MPC, not the MPC-DFT
- We take a semi-structural route: estimate a consumption function linking consumption to taxes, debt, wealth, expected income, and bond prices.

Why A Consumption Function?

Modest Ingredients

- household budget constraint
- Euler equation for bond holdings (with a non-Ricardian wedge)
- government budget constraint

Why A Consumption Function?

Modest Ingredients

- household budget constraint
- Euler equation for bond holdings (with a non-Ricardian wedge)
- government budget constraint

- Many potential sources of non-Ricardian behavior map to the same consumption function
- Clearest case: households have non-rational expectations over future taxes (internalize share θ only)
- Consumption function coefficients identify θ (but not with OLS...)

Our Approach

1. Derive the behavioral consumption function from budget constraints + Euler equation
2. Estimate with **B-HIVE** the **B**ayesian **H**ybrid **IV** **E**stimator, using macro structural shocks as external instruments to address endogeneity
   - Consumption function has many endogenous regressors
   - Instruments cover different samples, state space methods needed
   - Priors help with weak identification, theoretical constraints
3. Feed the estimate into a general equilibrium model to quantify the consequences of public borrowing

Our Approach

1. Derive the behavioral consumption function from budget constraints + Euler equation
2. Estimate with **B-HIVE** the **B**ayesian **H**ybrid **IV** **E**stimator, using macro structural shocks as external instruments to address endogeneity
   - Consumption function has many endogenous regressors
   - Instruments cover different samples, state space methods needed
   - Priors help with weak identification, theoretical constraints
3. Feed the estimate into a general equilibrium model to quantify the consequences of public borrowing

Goal

Estimate non-Ricardian behavior without taking a stance on a fully specified equilibrium model

Consumption Function

$$c_t = \phi_0 + \phi_n n_{t-1} + \phi_b b_{t-1} + \phi_\tau \tau_t + \tilde{\phi}_y v_t^y + \tilde{\phi}_g v_t^g + \tilde{\phi}_q v_t^q + \zeta_t$$

- n_{t-1}: net worth, b_{t-1}: debt, τ_t: taxes.
- v_t^y, v_t^g, v_t^q: sums of discounted expected values of income, spending, and bond prices.
- $\phi_n = (1-\beta)$: the MPC
- $\phi_\tau = -(1-\beta)(1-\theta)$: the MPC-DFT
- $\theta = 1$: Ricardian. $\theta < 1$: taxes/transfers move consumption

Consumption Function

$$c_t = \phi_0 + \phi_n n_{t-1} + \phi_b b_{t-1} + \phi_\tau \tau_t + \tilde{\phi}_y v_t^y + \tilde{\phi}_g v_t^g + \tilde{\phi}_q v_t^q + \zeta_t$$

- n_{t-1}: net worth, b_{t-1}: debt, τ_t: taxes.
- v_t^y, v_t^g, v_t^q: sums of discounted expected values of income, spending, and bond prices.
- $\phi_n = (1-\beta)$: the MPC
- $\phi_\tau = -(1-\beta)(1-\theta)$: the MPC-DFT
- $\theta = 1$: Ricardian. $\theta < 1$: taxes/transfers move consumption

Why Micro Evidence Is Not Enough (Revisited)

Cross-sectional designs vary who gets the transfer, not the common future tax burden; that is absorbed in the time fixed effect, so they do not recover ϕ_τ.

What Can Generate This Consumption Function?

In The Paper, We Derive This Object Under Three Sets Of Assumptions

$$c_t = \cdots - (1-\beta)(1-\theta)\tau_t + \cdots$$

1. **Behavioral expectations over taxes** (baseline):

$$\tilde{\mathbb{E}}_t[\tilde{v}^{\tau}_{t+1}] = \theta \mathbb{E}_t[v^{\tau}_{t+1}]$$

What Can Generate This Consumption Function?

In The Paper, We Derive This Object Under Three Sets Of Assumptions

$$c_t = \cdots - (1-\beta)(1-\theta)\tau_t + \cdots$$

1. **Behavioral expectations over taxes** (baseline):
$$\tilde{\mathbb{E}}_t[\tilde{v}^{\tau}_{t+1}] = \theta \mathbb{E}_t[v^{\tau}_{t+1}]$$
2. **Rational expectations + discounting wedge**: household discount factor is $\beta\omega$ rather than β, and if
$$v^{\tau}_t = \rho_\tau v^{\tau}_{t-1} + \varepsilon^{\tau}_t, \qquad \theta = \omega \frac{1-\beta\rho_\tau}{1-\beta\omega\rho_\tau}$$
then the consumption function is isomorphic to the behavioral one.

What Can Generate This Consumption Function?

In The Paper, We Derive This Object Under Three Sets Of Assumptions

$$c_t = \cdots - (1-\beta)(1-\theta)\tau_t + \cdots$$

1. **Behavioral expectations over taxes** (baseline):

$$\tilde{\mathbb{E}}_t[\tilde{v}^\tau_{t+1}] = \theta \mathbb{E}_t[v^\tau_{t+1}]$$

2. **Rational expectations + discounting wedge**: household discount factor is $\beta\omega$ rather than β, and if

$$v^\tau_t = \rho_\tau v^\tau_{t-1} + \varepsilon^\tau_t, \qquad \theta = \omega \frac{1-\beta\rho_\tau}{1-\beta\omega\rho_\tau}$$

then the consumption function is isomorphic to the behavioral one.

3. **Two-agent / hand-to-mouth economy**: with a share $1-\lambda$ of constrained households,

$$c^C_t = y^C_t - \tau_t, \qquad \theta = 1 - \frac{1-\lambda}{1-\beta}$$

Why Some Obvious Approaches Miss ϕ_τ

1. **Full-equation OLS:** the demand shock ζ_t is omitted, but in general equilibrium it also moves observed RHS variables. So regressors are correlated with the error term, and OLS cannot isolate ϕ_τ. (Feldstein, 1982; Cardia, 1997)

Why Some Obvious Approaches Miss ϕ_τ

1. **Full-equation OLS:** the demand shock ζ_t is omitted, but in general equilibrium it also moves observed RHS variables. So regressors are correlated with the error term, and OLS cannot isolate ϕ_τ. (Feldstein, 1982; Cardia, 1997)
2. **Lagged aggregates as IV:** if ζ_t is persistent, lagged consumption, wealth, or asset prices inherit past ζ_t and remain correlated with today's residual. So the exclusion restriction fails. (Hayashi, 1982; Feldstein, 1982; Seater and Mariano, 1985)

Why Some Obvious Approaches Miss ϕ_τ

1. **Full-equation OLS:** the demand shock ζ_t is omitted, but in general equilibrium it also moves observed RHS variables. So regressors are correlated with the error term, and OLS cannot isolate ϕ_τ. (Feldstein, 1982; Cardia, 1997)
2. **Lagged aggregates as IV:** if ζ_t is persistent, lagged consumption, wealth, or asset prices inherit past ζ_t and remain correlated with today's residual. So the exclusion restriction fails. (Hayashi, 1982; Feldstein, 1982; Seater and Mariano, 1985)
3. **Regressing consumption on exogenous tax shocks:** this estimates the total equilibrium response to a tax change, not the direct tax coefficient ϕ_τ. Tax shocks also move income, returns, and asset prices. (Mertens and Ravn, 2012; Leeper, Richter, and Walker, 2012)

Takeaway

None of these approaches isolate the direct tax coefficient ϕ_τ.

Monte Carlo appendix

Barnichon-Mesters: Structural Shocks as IVs

- Key idea: use externally identified macro shocks to estimate one structural equation. Barnichon and Mesters (2020, QJE)
- For the consumption function, the instruments must move the endogenous RHS variables while remaining excluded from the residual ζ_t (EE wedge)
- Intuition: productivity, monetary, fiscal, oil, etc. shocks all shift expected income, returns, wealth, taxes, or bond prices

Identification Logic

Use shocks that are excluded from the consumption equation but informative about its endogenous regressors.

B-HIVE Estimation

$$c_t = \phi' X_t + \zeta_t$$
$$X_t \equiv [1, n_{t-1}, b_{t-1}, \tau_t, \ldots]'$$
$$X_t = \mu_X + A_1 X_{t-1} + \cdots + A_p X_{t-p} + G\varepsilon_t$$
$$w_t = \mu_w + M_X X_{t-1} + M\varepsilon_t + \eta_t$$

- estimate one structural equation
- use the VAR in X_t to compute expected present values
- observe noisy shock instruments w_t
- Bayesian state space handles missing data, weak-IV concerns

Method appendix

State-Space Representation

Discounted Sums From The VAR

$$v_t^j = s_j'(I_{mp} - \beta_j F)^{-1}\overline{X}_t$$

where F is the companion-form transition matrix and $\overline{X}_t$ stacks current and lagged values of X_t.

- The present values on the RHS of the consumption equation are functions of the VAR coefficients.
- That creates cross-equation restrictions: the structural equation and state dynamics are estimated jointly.
- The state-space likelihood handles missing observations and different sample sizes naturally.

State-Space Representation

Discounted Sums From The VAR

$$v_t^j = s_j'(I_{mp} - \beta_j F)^{-1}\overline{X}_t$$

where F is the companion-form transition matrix and $\overline{X}_t$ stacks current and lagged values of X_t.

- The present values on the RHS of the consumption equation are functions of the VAR coefficients.
- That creates cross-equation restrictions: the structural equation and state dynamics are estimated jointly.
- The state-space likelihood handles missing observations and different sample sizes naturally.

Benefits of Our Approach

- model-consistent present values
- noisy external instruments
- Bayesian regularization under weak IV
- Standard Bayesian VAR toolkit available

State-Space Representation

Discounted Sums From The VAR

$$v_t^j = s_j'(I_{mp} - \beta_j F)^{-1} \overline{X}_t$$

where F is the companion-form transition matrix and $\overline{X}_t$ stacks current and lagged values of X_t.

- The present values on the RHS of the consumption equation are functions of the VAR coefficients.
- That creates cross-equation restrictions: the structural equation and state dynamics are estimated jointly.
- The state-space likelihood handles missing observations and different sample sizes naturally.

Benefits of Our Approach

- model-consistent present values
- noisy external instruments
- Bayesian regularization under weak IV
- Standard Bayesian VAR toolkit available

Extensions

The same framework can be extended to non-Gaussianity, stochastic volatility, etc.

Data Part 1: Macro Time Series

Observables

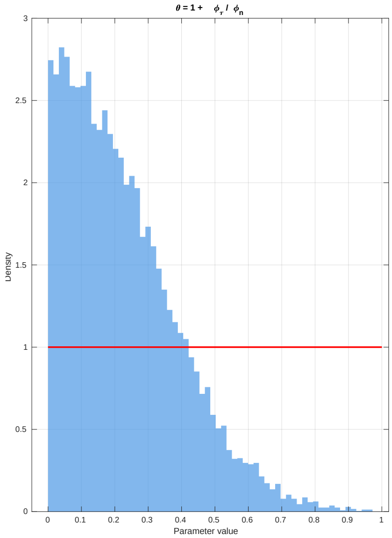
Quarterly U.S. data, 1969:Q1–2019:Q4.

- personal consumption
- personal taxes net of transfers
- personal income net of transfers
- household net worth
- market value of federal debt
- government expenditures
- 3-month Treasury bill yield
- SPF inflation forecasts
- real consol value (Aruoba term structure)
- survey income forecasts
- additional FRED-QD macro factors in X_t

Data Part 2: Macro Shocks (IVs)

Category	Examples used in the paper
Monetary policy	Jarocinski-Karadi, Miranda-Agrippino-Ricco, Bauer-Swanson, Aruoba-Drechsel
Government spending	Fisher-Peters, Ramey military news, Romer-Romer transfers, Fieldhouse-Mertens-Ravn housing, Fieldhouse-Mertens R&D
Tax / borrowing	Leeper-Walker-Yang, Mertens-Ravn, Lieb et al., Phillot
Technology	Fernald TFP, Miranda-Agrippino et al. patent news
Oil	Kilian, Kanzig, Baumeister-Hamilton
Other	Piffer-Podstawski uncertainty, Kim et al. severe weather, Chahrour-Jurado noise, Adams-Barrett inflation-expectations

Main Result I: Households Are Far From Ricardian



Behavioral Attenuation

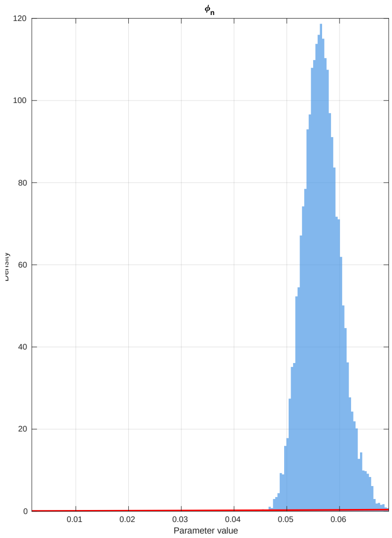
$$\theta = 0.19$$

90% credible interval:

$$[0.02,\ 0.54]$$

- $\theta = 1$ is the Ricardian benchmark.
- The posterior mass lies far below that benchmark.
- Baseline interpretation: households internalize only about one-fifth of future taxes.

Main Result II: The Generic MPC Is Small But Precisely Estimated



Quarterly MPC

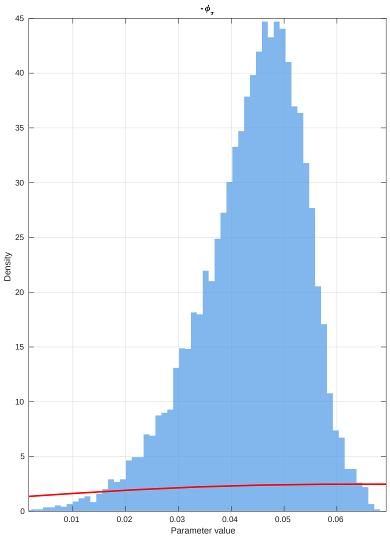
$$\text{MPC} = 0.056$$

90% credible interval:

$$[0.051,\ 0.063]$$

- The aggregate marginal propensity to consume out of net worth
- Prior is flat, posterior is tight
- Empirical estimates for MPCs out of illiquid wealth: 3%-7%

Main Result III: Debt-Financed Transfers Move Consumption



MPC Out Of Debt-Financed Transfers

$$\text{MPC-DFT} = 0.045$$

90% CI: [0.025, 0.058]

$$\text{MPC-DFT} = -\phi_\tau = (1-\beta)(1-\theta)$$

- Using $\phi_n = \text{MPC}$, we infer $\theta = 1 + \phi_\tau / \phi_n$
- Debt-financed transfers clearly increase consumption

Robustness: The Answer Stays Non-Ricardian

Selected Robustness Checks

Specification	θ	MPC-DFT
Baseline	0.194	0.045
Non-durable consumption	0.130	0.050
Survey-based forecasts	0.231	0.007
Non-separable utility	0.096	0.059
HtM income shares	0.123	0.066
Six selected instruments	0.174	0.027

- Each check targets a specific concern: richer expectations, labor in the SDF, HtM households, consumption measurement, or a leaner instrument set. Full table

From The Estimate To General Equilibrium

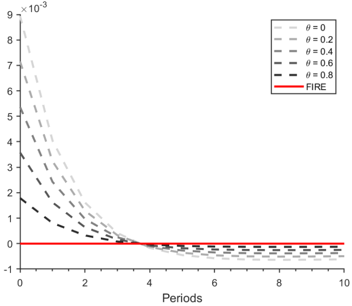
- Embed the estimated consumption function in a simple closed-economy RBC model with capital.
- A debt-financed tax cut makes households feel richer when $\theta < 1$.
- So consumption rises on impact, saving falls, and investment is crowded out.
- This is the equilibrium implication the reduced-form estimate is pointing to.

Mechanism

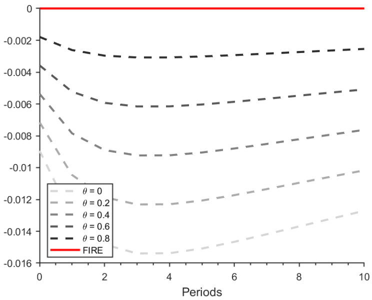
lower taxes today $\Rightarrow$ higher perceived wealth $\Rightarrow$ higher c_t $\Rightarrow$ lower i_t $\Rightarrow$ lower k_{t+1}

Equilibrium Implication: Government Debt Crowds Out Capital

Consumption response



Capital response



- Negative tax shock = debt-financed transfer.
- The smaller is θ, the stronger is the rise in consumption and the fall in capital.
- Quantitatively, our estimate implies substantial crowding out of private investment.

Takeaways

- The economic question is old, but the object we need is clear: the aggregate marginal propensity to consume out of debt-financed transfers
- Micro data are not enough: we need aggregate time series!
- Estimating the consumption function is a semi-structural strategy
- Barnichon-Mesters' insight: use other structural shocks to identify the equation
- Empirically, households are far from Ricardian: baseline $\theta \approx 0.19$.
- Combination of discounting wedges and non-rational expectations can rationalize
- In GE, non-Ricardianism implies that government borrowing crowds out capital.

Method Monte Carlo Mechanisms

Appendix: Full Consumption Function

$$c_t = (1-\beta)\Big(n_{t-1} - \theta b_{t-1} + \tilde{v}_t^y - (1-\theta)\tau_t - \theta v_t^g + \theta \bar{B} v_t^q\Big) + \Big(\frac{\bar{C}}{\gamma} - (1-\beta)\bar{B}\Big)\tilde{v}_t^q + \zeta_t$$

- $(1-\theta)\tau_t$ is the direct non-Ricardian tax term.
- Behavioral expectations motivate this mapping directly.
- Discounting wedges generate an isomorphic consumption function in the paper.
- HtM or liquidity-friction models imply a very similar aggregate equation in which current taxes still matter.

Back

Appendix: B-HIVE In One Slide

$$
\begin{aligned}
c_t &= \phi_0 + \phi_n n_{t-1} + \phi_b b_{t-1} + \phi_\tau \tau_t + \sum_{j\in\{y,g,q\}} \tilde{\phi}_j v_t^j + \zeta_t \\
X_t &= \mu_X + A_1 X_{t-1} + \cdots + A_p X_{t-p} + G\varepsilon_t \\
w_t &= \mu_w + M_X X_{t-1} + M\varepsilon_t + \eta_t
\end{aligned}
$$

- The VAR for X_t pins down the discounted expected-value terms.
- The measurement equation links observed shock series w_t to latent structural shocks ε_t.
- Bayesian inference helps with weak instruments and lets us impose economically natural bounds such as $0 \leq \theta \leq 1$.
- State-space methods naturally handle missing instruments and mixed sample coverage.

Back

Appendix: Full Robustness Table

Specification	θ	MPC	MPC-DFT
Baseline	0.194	0.056	0.045
Non-durable consumption	0.130	0.058	0.050
Survey-based forecasts	0.231	0.009	0.007
Non-separable utility	0.096	0.065	0.059
Variable distortionary taxes	0.597	0.058	0.023
HtM income shares	0.123	0.076	0.066
Including SVAR instruments	0.123	0.063	0.055
Six selected instruments	0.174	0.033	0.027

Back

Appendix: Non-Rational Expectations Vs. Discounting

How The Decomposition Works

Under additional assumptions on tax dynamics, β

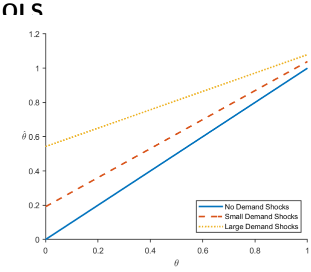
$$\phi_n = 1 - \beta\omega, \qquad -\phi_\tau = (1 - \beta\omega)(1 - \theta_{cd})$$

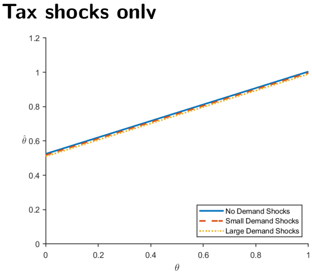
so two reduced-form moments pin down the discounting wedge ω and expectations wedge λ

Mechanism	λ	ω
Naive cognitive discounting	0.20	
Sophisticated cognitive discounting	0.90	
Discounting wedge + rational		0.90
Discounting wedge + naive C.D.	0.59	0.95
Discounting wedge + sophisticated C.D.	0.95	0.95

- Matching both MPC and MPC-DFT requires both $\lambda < 1$, $\omega < 1$ Back

Appendix: Monte Carlo I

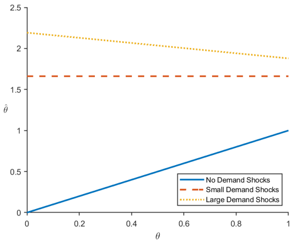




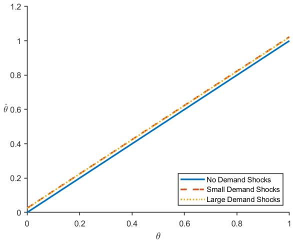
- OLS is biased when demand shocks move both consumption and endogenous regressors.
- "Just use tax shocks" is also biased because those shocks operate through other equilibrium margins.

Appendix: Monte Carlo II

Lagged variables as IVs



Structural-shock IVs



- Lagged aggregates fail once omitted shocks are persistent.
- Structural-shock IVs recover the correct coefficient.